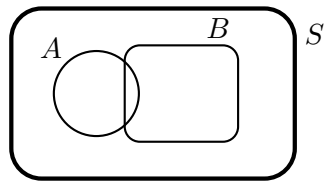
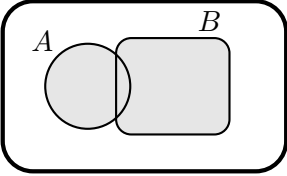
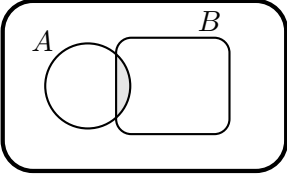
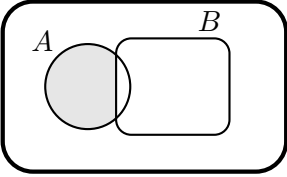
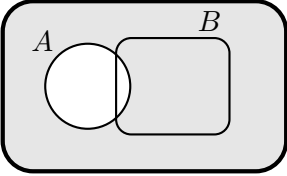
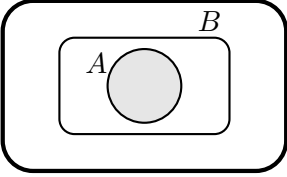


1. Let $A = \{12n \mid n \in \mathbb{Z}\}$, and let $B = \{k \in \mathbb{Z} \mid k \text{ is even}\}$, and let $C = \{3j \mid j \in \mathbb{Z}\}$. Show that $A \subseteq B \cap C$. That is, show that if $x \in A$, then $x \in B$ and $x \in C$.

Let's start with some notation and visual intuition. Generally speaking we'll be drawing an example as below, with A and B as subsets of a larger "universe" S .



Operation	Expression	Picture	Description
Union	$A \cup B$		$x \in A \cup B$ if $x \in A$ or $x \in B$.
Intersection	$A \cap B$		$x \in A \cap B$ if $x \in A$ and $x \in B$.
Set difference	$A \setminus B$		$x \in A \setminus B$ if $x \in A$ and $x \notin B$.
Complement	A' (or A^c or $\overline{A}$)		$x \in A'$ if $x \notin A$.
Subset	$A \subseteq B$ (or $B \supseteq A$)		$x \in A \implies x \in B$.

For each of the following, determine whether the statement is true for all sets A , B and C . If an equality fails, determine whether one or the other of the possible inclusions holds. Prove your assertions.

2. $(A \cap B) \cup (A \setminus B) = A$

3. $A \cup (B \setminus C) = (A \cup B) \setminus (A \cup C)$

4. Suppose $A = \{k \in \mathbb{Z} \mid 3 \mid k\}$ and $B = \{2k \mid k \in \mathbb{Z}\}$ are subsets of $\mathbb{Z}$. Describe (a) $(A \cup B)'$ and (b) $(A \cap B)'$.

The Joy of Sets – Solutions and Comments

Now, on to the solutions!

1. If $x \in A$, then $x = 12n$ for some integer n . Thus $x = 2(6n)$ (so $x \in B$) and $x = 3(4n)$ (so $x \in C$), and therefore $x \in B \cap C$. That is, $A \subseteq B \cap C$.
2. This seems to be true from a quick sketch (see Figure 1). As usual, to prove two sets are

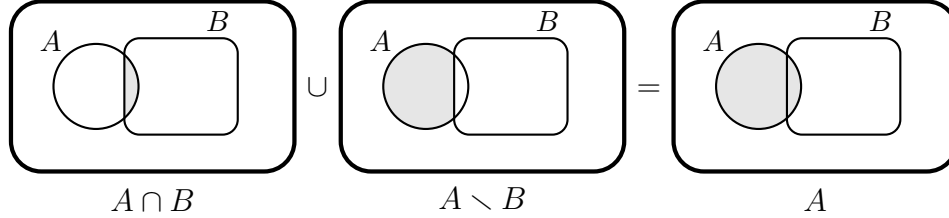


Figure 1: A quick sketch for the answer to Problem 2

equal, we prove each is a subset of the other. That is, to prove $S = T$, we prove that $S \subseteq T$ and $S \supseteq T$. We'll do this in parts:

(a) $(A \cap B) \cup (A \setminus B) \subseteq A$

If $x \in (A \cap B) \cup (A \setminus B)$, then either $x \in A \cap B$ (in which case $x \in A$) or $x \in A \setminus B$ (and again $x \in A$). Thus $x \in A$.

(b) $A \subseteq (A \cap B) \cup (A \setminus B)$

Suppose $x \in A$. If $x \in B$, then $x \in A \cap B$. If $x \notin B$, then $x \in A \setminus B$. Combining these, we see that $x \in (A \cap B) \cup (A \setminus B)$.

3. This is *not true*, since any element of A is included in the set on the left but is not included in the set on the right (it gets “subtracted away” when we remove $A \cup C$ from $A \cup B$). Only one inclusion holds:

Claim: $(A \cup B) \setminus (A \cup C) \subseteq A \cup (B \setminus C)$

Proof. Let $x \in (A \cup B) \setminus (A \cup C)$. Then $x \in A \cup B$ and $x \notin A \cup C$. That is, x is an element of either A or B **AND** x is an element of neither A nor C . That is, $x \notin A$, $x \in B$ and $x \notin C$. Thus $x \in B \setminus C$, so $x \in A \cup (B \setminus C)$. $\square$

4. Here $A \cup B$ can be described as all integers that are multiples of 2 or 3. The complement is therefore all integers that are *not* multiples of 2 or 3. Written out explicitly, we get

$$(A \cup B)' = \{\pm 1, \pm 5, \pm 7, \pm 11, \pm 13, \pm 17, \pm 19, \pm 23, \pm 25, \pm 29, \pm 31, \pm 35, \dots\}.$$

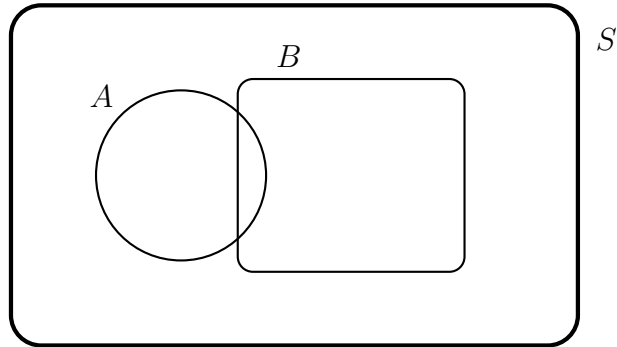
The intersection $A \cap B$ is the set of all integers that are multiples of *both* 2 and 3 – that is, it is the set of integers that are multiples of 6. The complement is thus all integers that are not divisible by 6:

$$(A \cap B)' = \{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 7, \pm 8, \pm 9, \pm 10, \pm 11, \pm 13, \dots\}.$$

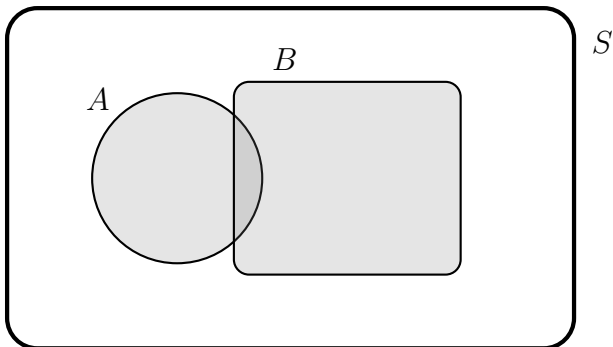
Part of the point of Problem 4 is to introduce the set-theoretic versions of [De Morgan's Laws](#):

$$(A \cup B)' = A' \cap B' \quad \text{and} \quad (A \cap B)' = A' \cup B'.$$

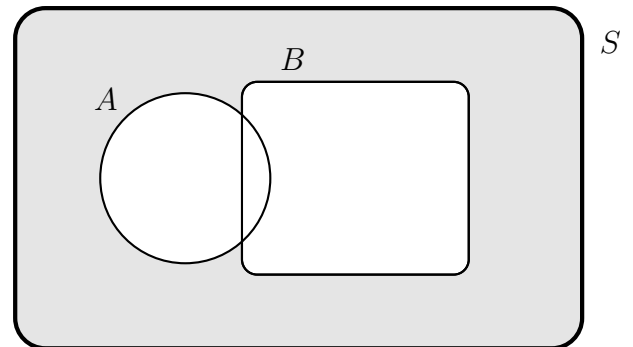
Here are the pictures to illustrate what's going on. Here's our entire set S , with the circular set A and the rectangular set B :



Then here are $A \cup B$ (left) and $A' \cap B'$ (right), showing that $(A \cup B)' = A' \cap B'$:



$A \cup B$ (shaded), so $(A \cup B)'$ is in white



$A' \cap B'$ is shaded

The other law is proved similarly; you can draw the diagram yourself.